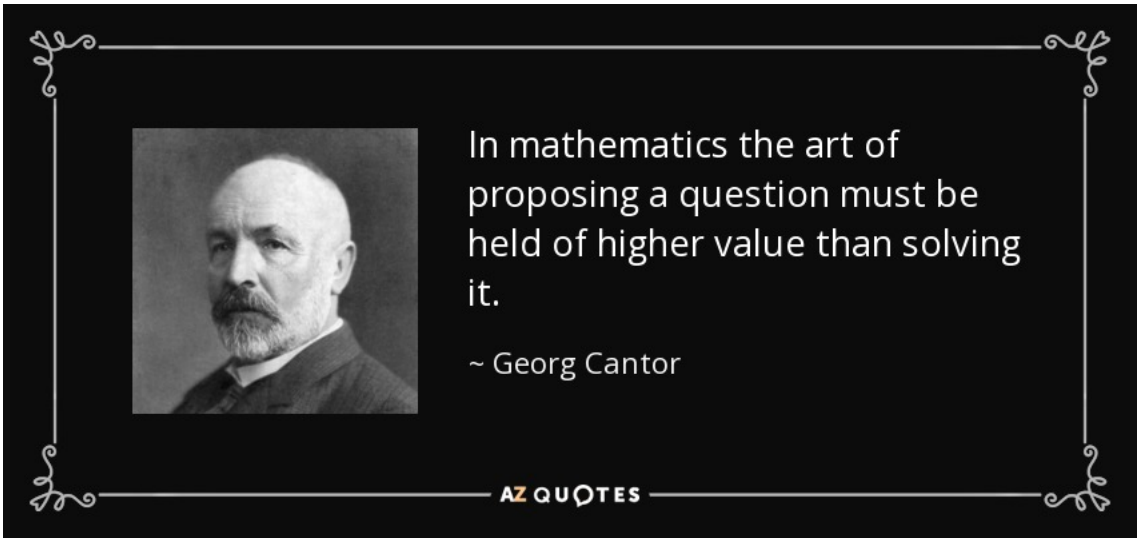
Chapter 1 - Logic and Proof

Note 4

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1.4 : Techniques of Proof: II

Mathematical theorems and proofs do not occur in isolation, but always in the context of some mathematical system. For example, when we discussed a conjecture related to prime numbers, the natural context of that discussion was the positive integers. When talking about odd and even numbers, the context was the set of all integers. Often a theorem will make no explicit reference to the mathematical system in which it is being proved; it must be inferred from the context. Usually, this causes no difficulty, but if there is a possibility of ambiguity, it's important to explicitly state the system being considered.

When dealing with quantified statements, it is particularly important to know exactly what system is being considered. For example, the statement $\forall\ x, \sqrt{x^2} = x$ is true in the context of the positive numbers, but is false when considering all real numbers.

Similarly, $\exists\ x \ni x^2 = 25$ and $x < 3$ is false for positive numbers and true for real numbers.

- To prove a universal statement $\forall\ x,\ p(x)$, we let x represent an arbitrary member from the system under consideration and then show that statement $p(x)$ is true. The only properties that we can use about x are those that apply to all the members of the system. For example, if the system consists of the integers, we cannot use the property that x is even, since this does not apply to all the integers.

- To prove an existential statement $\exists\, x \ni p(x)$, we have to prove that there is at least one member x in the system for which $p(x)$ is true. The most direct way of doing this is to construct (produce, guess, etc.) a specific x that has the required property.

Example 1 : To illustrate the process of writing a proof with quantifiers, consider the following:

> For every $\varepsilon > 0$ there exists a $\delta > 0$ such that
>
> $$1 - \delta < x < 1 + \delta \quad \text{implies that} \quad 5 - \varepsilon < 2x + 3 < 5 + \varepsilon.$$

We are asked to prove that something is true for each positive number ε. Thus we begin by letting ε be an arbitrary positive number. We need to use this ε to find a positive δ with the property that;

$$1 - \delta < x < 1 + \delta \quad \text{implies that} \quad 5 - \varepsilon < 2x + 3 < 5 + \varepsilon.$$

Let us begin with the consequent of the implication. We want to have

$$5 - \varepsilon < 2x + 3 < 5 + \varepsilon.$$

This will be true if

$$2 - \varepsilon < 2x < 2 + \varepsilon,$$

and this in turn will follow from

$$1 - \frac{\varepsilon}{2} < x < 1 + \frac{\varepsilon}{2}.$$

Thus we see that choosing $\delta = \varepsilon/2$ will meet the required condition. In writing down the proof in a formal manner we would simply set $\delta = \varepsilon/2$ and then show that this particular δ will work.

Proof. Given any $\varepsilon > 0$, let $\delta = \varepsilon/2$. Then δ is also positive and whenever

$$1 - \delta < x < 1 + \delta$$

we have

$$1 - \frac{\varepsilon}{2} < x < 1 + \frac{\varepsilon}{2},$$

so that

$$2 - \varepsilon < 2x < 2 + \varepsilon \quad \text{and} \quad 5 - \varepsilon < 2x + 3 < 5 + \varepsilon,$$

as required. □

In some situations it is possible to prove an existential statement in an indirect way without actually producing any specific member of the system. One indirect method is to use the **contrapositive** form of the implication and another is to use a proof by **contradiction**.

The two basic forms of a proof by contradiction are based on tautologies (f) and (g) in Example 12 in Note 3.

- Tautology (f) has the form $(\sim p \Longrightarrow c) \Longleftrightarrow p$. If we wish to conclude a statement p, we can do so by showing that the negation of p leads to a contradiction.
- Tautology (g) has the form $(p \Longrightarrow q) \Longleftrightarrow [\ (p \wedge \sim q) \Longrightarrow c\ ]$. If we wish to conclude that p implies q, we can do so by showing that p and not q leads to a contradiction.

In either case, the contradiction can involve part of the hypothesis or some other statement that is known to be true.

Example 2 : Use truth tables to verify that $(\sim p \Longrightarrow c) \Longleftrightarrow p$ and $[\ (p \wedge \sim q) \Longrightarrow c\ ] \Longleftrightarrow (p \Longrightarrow q)$ are tautologies.

Solution :

p	$(\sim p \Rightarrow c)$	$\Leftrightarrow$	p
T	F T F	T	T
F	T F F	T	F

p	q	$[(p \wedge \sim q) \Rightarrow c]$	$\Leftrightarrow$	$(p \Rightarrow q)$
T	T	T F F T F	T	T
T	F	T T T F F	T	F
F	T	F F F T F	T	T
F	F	F F T T F	T	T

Example 3 : To illustrate an indirect proof of an existential statement, consider the following:

> Let f be an integrable function. If $\int_0^1 f(x)\,dx \neq 0$, then there exists a point x in the interval $[0, 1]$ such that $f(x) \neq 0$.

Symbolically, we have $p \Longrightarrow q$, where

$$p : \int_0^1 f(x)\ dx \neq 0$$

$$q : \exists\ x \text{ in } [0,1] \ni f(x) \neq 0$$

The contrapositive implication, $\sim q \implies \sim p$, can be written as

$$\text{If for every } \ x \text{ in } [0,1],\ f(x) = 0, \text{ then } \int_0^1 f(x) = 0.$$

Example 4 : If a finite sum of real numbers equals a non-zero number, i.e., $a_1 + a_2 + \cdots + a_n \neq 0$, then there exists at least one term a_i such that $a_i \neq 0$.

Proof : Homework !

To illustrate a proof by contradiction, consider Examples 5 and 6.

Example 5 : Let x be a real number. If $x > 0$, then $1/x > 0$. Symbolically, we have $p \Longrightarrow q$, where $p : x > 0$ and $q : \dfrac{1}{x} > 0$.

Proof.

- Tautology (g) in Example 12 in Note 3 says that $p \Rightarrow q$ is equivalent to $(\ p \wedge \sim q\ ) \Rightarrow c$.
- Suppose $x > 0$ and $1/x \leq 0$.
- Since $x > 0$, we can multiply both sides of the inequality $1/x \leq 0$ by x to obtain

$$(x)\left(\frac{1}{x}\right) \leq (x)(0)$$

- But $(x)(1/x) = 1$ and $(x)(0) = 0$, so we have $1 \leq 0$, a contradiction to the (presumably known) fact that $1 > 0$.
- Having shown that $p \wedge \sim q$ leads to a contradiction, we conclude that $p \Longrightarrow q$.

□

Example 6 : Prove that $\sqrt{2}$ is irrational.

Proof : Homework !

Another tautology in Example 12 in Note 3 that deserves special attention is statement (q):

$$[(p \vee q) \Longrightarrow r] \Longleftrightarrow [(p \Longrightarrow r) \wedge (q \Longrightarrow r)].$$

Some proofs naturally divide themselves into the consideration of two (or more) cases. For example, integers are either odd or even. Real numbers are positive, negative, or zero. It may be that different arguments are required for each case. It is tautology (q) that shows us how to combine the cases.

Example 7 : Suppose we wish to prove that if x is a real number, then $x \leq |x|$. Symbolically, we have $s \Longrightarrow r$, where $s : x$ is a real number and $r : x \leq |x|$.

First, we recall the definition of absolute value:

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

Since this definition is divided into two parts, it is natural to divide our proof into two cases. Thus statement s is replaced by the equivalent disjunction $p \vee q$, where $p : x \geq 0$ and $q : x < 0$.

Our theorem now is to prove $(p \vee q) \Rightarrow r$, and we do this by showing that $(p \Rightarrow r) \wedge (q \Rightarrow r)$. The actual proof could be written as follows :

Proof.

- Let x be an arbitrary real number. Then $x \geq 0$ or $x < 0$.
- **Case I :** If $x \geq 0$, then by definition, $x = |x|$.
- **Case II :** If $x < 0$, then $-x > 0$, so that $x < 0 < -x = |x|$.
- Thus, in either case, we have $x \leq |x|$.

□

Example 8 : Show that $|xy| = |x||y|$, where x and y are real numbers.

Proof : Homework !

Example 9 : In proving a theorem that relates to factoring positive integers greater than 1, what two cases might reasonably be considered?

Solution : The positive integers greater than 1 are either prime or composite. They are also either odd or even. Either way of separating the integers into two cases could be reasonable, depending on the context.

An alternative form of proof by cases arises when the consequent of an implication involves a disjunction. In this situation, tautology (p) of Example 12 in Note 3 is often helpful:

$$[p \Longrightarrow (q \vee r)] \Longleftrightarrow [(p \wedge \sim q) \Longrightarrow r].$$

Example 10 : If the sum of a real number with itself is equal to its square, then the number is 0 or 2.

In symbols we have $p \Rightarrow (q \vee r)$, where $p : x + x = x^2$, $q : x = 0$, and $r : x = 2$. We shall show that $p \wedge (\sim q) \Rightarrow r$.

Proof. Suppose that $x + x = x^2$ and $x \neq 0$. Then $2x = x^2$ and since $x \neq 0$, we can divide by x to obtain $2 = x$. □

Example 11 : Suppose that you wish to prove the statement "If B is both open and closed, then $B = \varnothing$ or $B = X$". Use tautology (p) of Example 3.12 to state *two* different equivalent statements that could be proved instead.

Solution : If B is both open and closed and $B \neq \varnothing$, then $B = X$. If B is both open and closed and $B \neq X$, then $B = \varnothing$.

We have now discussed three ways of proving a statement of the form $p \Longrightarrow q$:

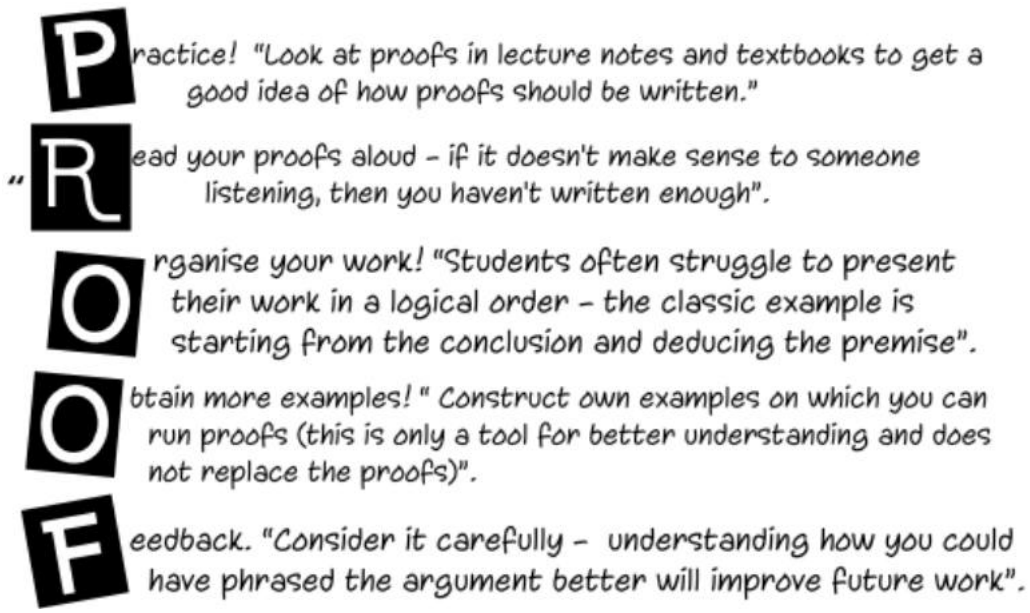
(i). Assume statement p and deduce statement q.

(ii). Assume $\sim q$ and deduce $\sim p$. (Prove the contrapositive.)

(iii). Assume both p and $\sim q$ and deduce a contradiction.

Epilogue : You will have the opportunity to read and write a great many proofs. Make the most of it! When you read a proof, analyze its structure. See what tautologies, if any, have been used. Note the important role that definitions play. Often a proof will be little more than unraveling definitions and applying them to specific cases. From time to time we shall point out the method to be used in a proof to help you see the structure that we shall be following.

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Practice! "Look at proofs in lecture notes and textbooks to get a good idea of how proofs should be written."
"Read your proofs aloud – if it doesn't make sense to someone listening, then you haven't written enough".
Organise your work! "Students often struggle to present their work in a logical order – the classic example is starting from the conclusion and deducing the premise".
Obtain more examples! " Construct own examples on which you can run proofs (this is only a tool for better understanding and does not replace the proofs)".
Feedback. "Consider it carefully – understanding how you could have phrased the argument better will improve future work".